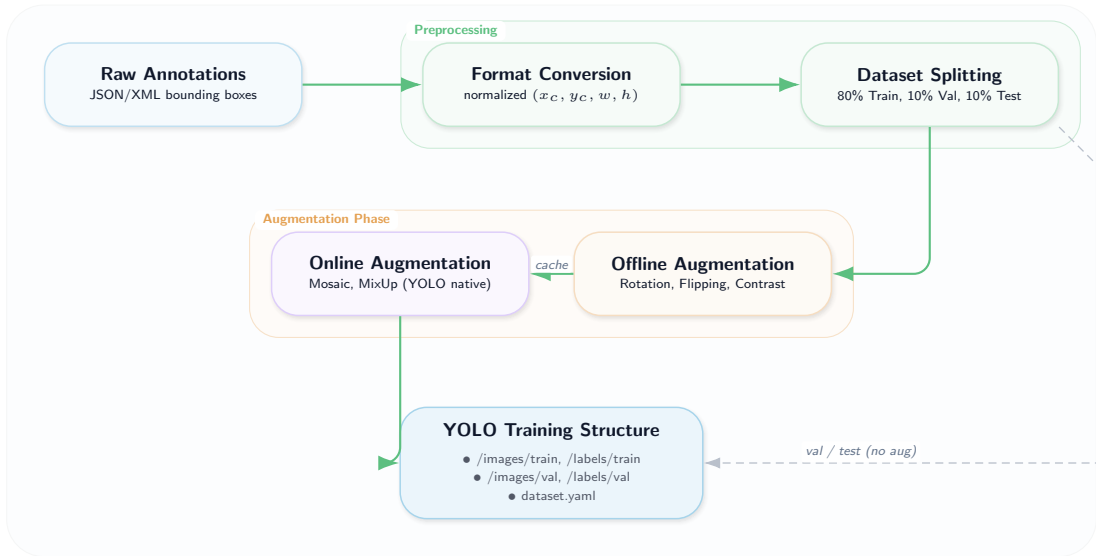




Augmentation Strategy — Offline augmentation expands the static dataset to combat class imbalance. Online augmentation (Mosaic, MixUp) is applied dynamically during YOLO training by the dataloader to improve model generalization and detection robustness on variable scales.